Hence, $\vdash \alpha \Leftrightarrow \beta$. ■

Definition: The ***converse*** of $(\alpha \Rightarrow \beta)$ is the wff $(\beta \Rightarrow \alpha)$. Therefore, to prove that $(\alpha \Leftrightarrow \beta)$ is a theorem, we may prove that both $(\alpha \Rightarrow \beta)$ and its converse, $(\beta \Rightarrow \alpha)$, are theorems.

Example 1.7.14 Prove that $\vdash \alpha \vee (\alpha \wedge \sim \beta) \Leftrightarrow \alpha$.

We shall prove that $\vdash \alpha \vee (\alpha \wedge \sim \beta) \Rightarrow \alpha$ and $\vdash \alpha \Rightarrow \alpha \vee (\alpha \wedge \sim \beta)$.

(i) Prove that $\vdash \alpha \vee (\alpha \wedge \sim \beta) \Rightarrow \alpha$.

(Direct Proof)

1.	$\alpha \vee (\alpha \wedge \sim \beta)$	Hypothesis
2.	$(\alpha \vee \alpha) \wedge (\alpha \vee \sim \beta)$	1, E14
3.	$(\alpha \vee \alpha)$	2, I2
4.	α	3, E4

Hence, $\vdash \alpha \vee (\alpha \wedge \sim \beta) \Rightarrow \alpha$.

(ii) Prove that $\vdash \alpha \Rightarrow \alpha \vee (\alpha \wedge \sim \beta)$.

(Direct Proof)

1.	α	Hypothesis
2.	$\alpha \vee (\alpha \wedge \sim \beta)$	1, I1

Hence, $\vdash \alpha \Rightarrow \alpha \vee (\alpha \wedge \sim \beta)$

Combining the results of (i) and (ii), we have $\vdash \alpha \vee (\alpha \wedge \sim \beta) \Leftrightarrow \alpha$. ■

The Soundness and Completeness of Propositional Logic

A formal proof theory of a logic system is ***sound*** if every syntactic construct derivable from a formal proof possesses the desired semantic property of being true in any circumstance.

A formal proof theory of a logic system is ***complete*** if every syntactic construct that possesses the desired semantic property of being true in any circumstance has a formal proof.

In the context of the formal proof theory developed in the previous section, a syntactic construct derivable from a formal proof refers to a theorem, and a syntactic construct possesses the desired semantic property of being true in any circumstance refers to a tautology. Therefore, the formal proof theory is sound if every theorem is a tautology. It is complete if every tautology is a theorem.

Soundness is a must if the formal proof theory is to be useful, for we do not want to derive a theorem which may be false. Completeness, on the other hand, although it is nice to have, is not a must. As a matter of fact, it has been proven that not all branches of mathematical logic are complete.

A question that naturally arises is: "Is the formal proof theory in Propositional logic sound and complete?" This is answered affirmatively in the following theorem.

Theorem 1.7.1: *(The Soundness and Completeness Theorem of Propositional Logic)*

A wff α is a theorem iff α is a tautology.

Proof: Omitted. ■

The following are some useful theorems.

Theorem 1.7.2:

(i) $\vdash \sim false \Leftrightarrow true$
$\vdash \sim true \Leftrightarrow false$
(ii) $\vdash (\alpha \Leftrightarrow \beta) \Leftrightarrow (\beta \Leftrightarrow \alpha)$
(iii) $\vdash (\alpha \Leftrightarrow \beta) \Leftrightarrow (\sim \alpha \Leftrightarrow \sim \beta)$
(iv) $\vdash (\alpha \Leftrightarrow \beta) \wedge (\beta \Leftrightarrow \gamma) \Rightarrow (\alpha \Leftrightarrow \gamma)$
(v) $\vdash (\alpha \Rightarrow (\beta \Rightarrow \gamma)) \Leftrightarrow (\alpha \wedge \beta \Rightarrow \gamma)$

Proof: (Left as exercises). ■

Owing to Theorem 1.7.1, each of the theorems in Theorem 1.7.2 is a tautology. As a result, each of them gives rise to an equivalence or an inference rule. Substitution can thus be applied using the equivalence. As an example, consider the wff, $p \vee (q \Leftrightarrow r)$. By substitution using Theorem 1.7.2(iii) (which gives rise to the equivalence $(\alpha \Leftrightarrow \beta) \equiv (\sim \alpha \Leftrightarrow \sim \beta)$), we obtain the wff, $p \vee (\sim q \Leftrightarrow \sim r)$. This remark applies equally well to the rest of these lecture notes.

Examples

(***Remark***: In the following examples, we will not state the use of the substitution rule explicitly in the proofs).

Example 1.7.15 Prove that $\vdash ((p \Rightarrow q) \Rightarrow q) \Rightarrow (p \vee q)$.

Solution: (Direct proof)

1.	$((p \Rightarrow q) \Rightarrow q)$	Hypothesis
2.	$((\sim p \vee q) \Rightarrow q)$	1, E18
3.	$\sim (\sim p \vee q) \vee q$	2, E18
4.	$(\sim\sim p \wedge \sim q) \vee q$	3, E17
5.	$(p \wedge \sim q) \vee q$	4, E15
6.	$q \vee (p \wedge \sim q)$	5, E10
7.	$(q \vee p) \wedge (q \vee \sim q)$	6, E14
8.	$(q \vee p)$	7, I2
9.	$(p \vee q)$	8, E10

Hence, $\vdash ((p \Rightarrow q) \Rightarrow q) \Rightarrow (p \vee q)$ ■

Example 1.7.16 Prove that $p \Rightarrow q,\ r \Rightarrow (\sim q \vee s),\ t \Rightarrow (r \vee u),\ s \Rightarrow \sim q \vdash t \Rightarrow (p \Rightarrow u)$.

Solution: (Direct proof)

1.	t	Hypothesis
2.	$p \Rightarrow q$	from Γ
3.	$r \Rightarrow (\sim q \vee s)$	from Γ
4.	$t \Rightarrow (r \vee u)$	from Γ
5.	$s \Rightarrow \sim q$	from Γ
6.	$(r \vee u)$	1,4, I3
7.	$(u \vee r)$	6, E10

8. $(\sim\sim u \vee r)$ 7, E15
9. $(\sim u \Rightarrow r)$ 8, E18
10. $\sim u \Rightarrow (\sim q \vee s)$ 9,3, I5
11. $\sim\sim u \vee (\sim q \vee s)$ 10, E18
12. $(\sim\sim u \vee \sim q) \vee s$ 11, E12
13. $\sim (\sim u \wedge q) \vee s$ 12, E16
14. $(\sim u \wedge q) \Rightarrow s$ 13, E18
15. $(\sim u \wedge q) \Rightarrow \sim q$ 14,5, I5
16. $\sim (\sim u \wedge q) \vee \sim q$ 15, E18
17. $(\sim\sim u \vee \sim q) \vee \sim q$ 16, E16
18. $(u \vee \sim q) \vee \sim q$ 17, E15
19. $u \vee (\sim q \vee \sim q)$ 18, E12
20. $u \vee \sim q$ 19, E4
21. $\sim q \vee u$ 20, E10
22. $q \Rightarrow u$ 21, E18
23. $p \Rightarrow u$ 2,22, I5

Hence, $p \Rightarrow q,\ r \Rightarrow (\sim q \vee s),\ t \Rightarrow (r \vee u),\ s \Rightarrow \sim q \vdash t \Rightarrow (p \Rightarrow u)$ ■

The conclusion in the above example has the form $\alpha \Rightarrow (\beta \Rightarrow \gamma)$. In general, to prove a theorem of the form $\Gamma \vdash \alpha \Rightarrow (\beta \Rightarrow \gamma)$, we may prove that $\Gamma, \alpha, \beta \vdash \gamma$. This is because if $\Gamma, \alpha, \beta \vdash \gamma$, then by applying the deduction theorem twice, we have $\Gamma \vdash \alpha \Rightarrow (\beta \Rightarrow \gamma)$. Actually, what we have is a direct proof nested within another direct proof. Therefore, the last example could be redone as follows.

Prove that $\vdash p \Rightarrow q,\ r \Rightarrow (\sim q \vee s),\ t \Rightarrow (r \vee u),\ s \Rightarrow \sim q,\ t,\ p \vdash u$

1. t from Γ
2. p from Γ
3. $p \Rightarrow q$ from Γ
4. $r \Rightarrow (\sim q \vee s)$ from Γ
5. $t \Rightarrow (r \vee u)$ from Γ
6. $s \Rightarrow \sim q$ from Γ
7. q 2,3, I3
8. $\sim\sim q$ 7, E15
9. $\sim s$ 8,6, I4
10. $\sim\sim q \wedge \sim s$ 8,9, I6
11. $\sim (\sim q \vee s)$ 10, E17
12. $\sim r$ 11,4, I4
13. $(r \vee u)$ 1,5, I3
14. $(\sim\sim r \vee u)$ 13, E15
15. $\sim r \Rightarrow u$ 14, E18
16. u 12,15, I3

Hence, $p \Rightarrow q,\ r \Rightarrow (\sim q \vee s),\ t \Rightarrow (r \vee u),\ s \Rightarrow \sim q,\ t,\ p \vdash u$.

By the Deduction Theorem, we have $p \Rightarrow q,\ r \Rightarrow (\sim q \vee s),\ t \Rightarrow (r \vee u),\ s \Rightarrow \sim q,\ t \vdash (p \Rightarrow u)$.

By the Deduction Theorem again, we obtain:

$p \Rightarrow q,\ r \Rightarrow (\sim q \vee s),\ t \Rightarrow (r \vee u),\ s \Rightarrow \sim q \vdash t \Rightarrow (p \Rightarrow u)$ ■.

Example 1.7.17 Prove that $q \Rightarrow t,\ s \Rightarrow r,\ q \vee s \vdash\ t \vee r$.
Solution:

1.	$q \Rightarrow t$	from Γ
2.	$s \Rightarrow r$	from Γ
3.	$q \vee s$	from Γ
4.	$s \vee q$	3, E10
5.	$\sim\sim s \vee q$	4, E15
6.	$\sim s \Rightarrow q$	5, E18
7.	$\sim s \Rightarrow t$	6,1, I5
8.	$\sim r \Rightarrow \sim s$	2,E19
9.	$\sim r \Rightarrow t$	8,7,I5
10.	$\sim\sim r \vee t$	9,E18
11.	$r \vee t$	10,E15
12.	$t \vee r$	11,E10

Hence, $q \Rightarrow t,\ s \Rightarrow r,\ q \vee s \vdash\ t \vee r$ ■

1.8 Applications

Logic can be used to test the validity of an argument. An argument is the assertion that a proposition q follows from a number of other propositions $p_1, p_2, \cdots, p_k$ i.e. $p_1, p_2, \cdots, p_k$ logically implies q. Therefore, to prove that the argument is valid, we may prove that $p_1, p_2, \cdots, p_k \vdash q$ or equivalently $\vdash (p_1 \wedge p_2 \wedge \cdots \wedge p_k) \Rightarrow q$.

Example 1.8.18 If he plays basketball, then he prepares to play in the NBA. If he plays ice hockey, then he prepares to play in the NHL. If he prepares to play in the NBA or the NHL, then his time is not wasted. His time is wasted. Therefore, he plays neither basketball nor ice hockey.

Solution:

Let B denote he plays basketball,
H denote he plays ice hockey,
N denote he prepares to play in the NBA,
L denotes he prepares to play in the NHL,
W denotes his time is wasted.

The first sentence is of the form "if B then N" and can thus be translated into $B \Rightarrow N$

The second sentence is of the form "if H then L" and can thus be translated into $H \Rightarrow L$

The third sentence is of the form "if (N or L) then *not* W" and can thus be translated into $(N \vee L) \Rightarrow \sim W$

The fourth sentence is easily translated into W.

The conclusion is of the form " *(not B) and (not H)*" and can thus be translated into $\sim B \wedge \sim H$.

We thus have:

$P_1 : B \Rightarrow N$
$P_2 : H \Rightarrow L$
$P_3 : (N \vee L) \Rightarrow \sim W$
$P_4 : W$
$C : \sim B \wedge \sim H$

and our task is to prove $P_1, P_2, P_3, P_4 \vdash C$.

1.	$B \Rightarrow N$	from Γ
2.	$H \Rightarrow L$	from Γ
3.	$(N \vee L) \Rightarrow \sim W$	from Γ
4.	W	from Γ
5.	$\sim\sim W$	4, E15
6.	$\sim (N \vee L)$	5,3, I4
7.	$\sim N \wedge \sim L$	6, E17
8.	$\sim N$	7, I2
9.	$\sim L \wedge \sim N$	7, E9
10.	$\sim L$	9, I2
11.	$\sim B$	8,1, I4
12.	$\sim H$	10,2, I4
13.	$\sim B \wedge \sim H$	11,12, I6

Hence, $P_1, P_2, P_3, P_4 \vdash C$. ■

Example 1.8.19 Given that if the board and the union refuse to negotiate, then the strike will not be over unless it lasts more than one year and the chairman of the board resigns. Will the strike not be over if the board and the union refuse to negotiate and the strike has just started?

Solution: The conclusion is: "The strike will not be over." A proof of its validity is given below.

Let N stands for the board and the union refuse to negotiate
S stands for the strike will be over
R stands for the chairman of the board resigns
O stands for the strike lasts more than one year

In the first sentence, 'the strike will not be over unless it lasts more than one year and the chairman of the board resigns' has the form: *not* S unless (O and R) and can thus be translated into $S \Rightarrow (O \wedge R)$.

The entire sentence can thus be translated into $N \Rightarrow (S \Rightarrow (O \wedge R))$.

The sentence: "the board and the union refuse to negotiate and the strike just starts" can be translated into $N \wedge \sim O$. Note that as the strike has just started, it cannot be true that it has lasted for more than one year. Therefore, the sentence "the strike has just started" is translated into $\sim O$. The conclusion is translated into $\sim S$. We thus have;

$P_1 : N \Rightarrow (S \Rightarrow (O \wedge R))$
$P_2 : N \wedge \sim O$
$C : \sim S$

We are to prove that: $P_1, P_2 \vdash C$.

1.	$N \Rightarrow (S \Rightarrow (O \wedge R))$	from Γ
2.	$N \wedge \sim O$	from Γ
3.	N	2, I2
4.	$(S \Rightarrow (O \wedge R))$	3,1, I3
5.	$\sim O \wedge N$	2, E9
6.	$\sim O$	5, I2
7.	$\sim O \vee \sim R$	6, I1
8.	$\sim (O \wedge R)$	7, E16

9. $\sim S$ 8,4, I4

Hence, $P_1, P_2 \vdash C$. ■

Example 1.8.20 If the Red Wings win then Detroit will celebrate. If the Rangers win, then New York will celebrate. Either the Red Wings win or the Ranger win. If the Red Wings win, then New York will not celebrate, and if the Ranger win, then Detroit will not celebrate. Therefore, Detroit wll celebrate if and only if New York will not celebrate.

Solution:

Let W denote "the Red Wings win,"
R denote "the Rangers win,"
D denote "Detroit will celebrate,"
N denotes "New York will celebrate."

The sentence: "If the Red Wings win then Detroit will celebrate" can be translated into $W \Rightarrow D$

The sentence: "If the Rangers win, then New York will celebrate" can be translated into $R \Rightarrow N$

The sentence: "Either the Red Wings win or the Ranger win." can be translated into $(W \vee R) \wedge \sim (W \wedge R)$ because there can be only one winner.

The sentence: "If the Red Wings win, then New York will not celebrate, and if the Ranger win, then Detroit will not celebrate" can be translated into $(W \Rightarrow \sim N) \wedge (R \Rightarrow \sim D)$

The conclusion: "Detroit wll celebrate if and only if New York will not celebrate" can be translated into $D \Leftrightarrow \sim N$

We thus have:

$P_1 : W \Rightarrow D$
$P_2 : R \Rightarrow N$
$P_3 : (W \vee R) \wedge \sim (W \wedge R)$
$P_4 : (W \Rightarrow \sim N) \wedge (R \Rightarrow \sim D)$
$C : D \Leftrightarrow \sim N$

and we are to prove that $P_1, P_2, P_3, P_4 \vdash C$.

1. $W \Rightarrow D$ from Γ
2. $R \Rightarrow N$ from Γ
3. $(W \vee R) \wedge \sim (W \wedge R)$ from Γ
4. $(W \Rightarrow \sim N) \wedge (R \Rightarrow \sim D)$ from Γ
5. $(R \Rightarrow \sim D) \wedge (W \Rightarrow \sim N)$ 4, E9
6. $(R \Rightarrow \sim D)$ 5, I2
7. $(\sim\sim D \Rightarrow \sim R)$ 6, E19
8. $(D \Rightarrow \sim R)$ 7, E15
9. $(W \vee R)$ 3, I2
10. $R \vee W$ 9, E10
11. $\sim\sim R \vee W$ 10, E15
12. $(\sim R \Rightarrow W)$ 11, E18
13. $(D \Rightarrow W)$ 8,12, I5

14. $(W \Rightarrow \sim N)$ 4, I2
15. $(D \Rightarrow \sim N)$ 13,14, I5
16. $(\sim N \Rightarrow \sim R)$ 2, E19
17. $(\sim N \Rightarrow W)$ 16,12, I5
18. $(\sim N \Rightarrow D)$ 17,1, I5
19. $(D \Rightarrow \sim N) \wedge (\sim N \Rightarrow D)$ 15,18, I6
20. $D \Leftrightarrow \sim N$ 19, E20

Hence, $P_1, P_2, P_3, P_4 \vdash C$. ■

Example 1.8.21 If the driver was conscious and was aware of the speed at which he was driving, and the speedometer was functioning, then mechanical failure was responsible for the crash. Inspection of the wreckage shows that there was no mechanical failure and that the speedometer was functioning. Therefore, if the driver was conscious, then he wasn't aware of the speed at which he was driving.

Solution:

Let c denote the driver was conscious,
k denote the driver was aware of the speed at which he was driving,
a denote the speedometer was functioning,
f denote there was mechanical failure

The given argument can be translated into the following wffs.

$P_1 : (c \wedge k \wedge a) \Rightarrow f$
$P_2 : \sim f \wedge a$
$C : c \Rightarrow \sim k$

We are to prove that $P_1, P_2 \vdash C$.

1. $(c \wedge k \wedge a) \Rightarrow f$ from Γ
2. $\sim f \wedge a$ from Γ
3. $\sim f$ 2,I2
4. $\sim (c \wedge k \wedge a)$ 3,1,I4
5. $\sim (c \wedge k) \vee \sim a$ 4,E16
6. $\sim a \vee \sim (c \wedge k)$ 5,E10
7. $a \Rightarrow \sim (c \wedge k)$ 6,E18
8. $a \wedge \sim f$ 2,E9
9. a 8,I2
10. $\sim (c \wedge k)$ 9,7,I3
11. $\sim c \vee \sim k$ 10,E16
12. $c \Rightarrow \sim k$ 11,E18

Hence, $P_1, P_2 \vdash C$. ■

1.9 Exercises

1. Construct a truth table for each of the following wffs:
   a. $(((p \Rightarrow q) \Rightarrow q) \Rightarrow q)$
   b. $(((p \Rightarrow q) \Rightarrow (r \Rightarrow p)) \Rightarrow (r \Rightarrow p))$
   c. $(p \vee \sim (q \wedge r)) \Rightarrow ((p \Leftrightarrow r) \vee q)$
   d. $(p \vee q) \wedge (\sim p \vee \sim r)$

e. $(\sim r \wedge q) \vee (p \wedge q) \vee (\sim p \wedge \sim r)$
f. $(\sim p \vee q) \wedge (q \Rightarrow \sim r \wedge \sim p) \wedge (p \vee r)$
g. $(p \vee q) \wedge (p \Rightarrow r \wedge q) \wedge (q \Rightarrow \sim r \wedge p)$

2. Prove that each of the following wffs is a tautology with the truth table.
a. $((\alpha \Rightarrow \gamma) \wedge (\beta \Rightarrow \gamma)) \Rightarrow ((\alpha \vee \beta) \Rightarrow \gamma)$
b. $\beta \vee \sim (\alpha \Rightarrow (\beta \vee \gamma)) \vee (\alpha \Rightarrow \gamma)$
c. $\alpha \Rightarrow (\beta \Rightarrow \alpha)$
d. $(\alpha \Rightarrow (\beta \Rightarrow \gamma)) \Rightarrow ((\alpha \Rightarrow \beta) \Rightarrow (\alpha \Rightarrow \gamma))$
e. $(\sim\beta \Rightarrow \sim\alpha) \Rightarrow ((\sim\beta \Rightarrow \alpha) \Rightarrow \beta)$
f. $(\alpha \Rightarrow \beta) \vee (\beta \Rightarrow \alpha)$
g. $(\alpha \Rightarrow \beta) \Rightarrow ((\alpha \wedge \gamma) \Rightarrow (\beta \wedge \gamma))$
h. $(\alpha \Rightarrow (\beta \wedge \sim\beta)) \Rightarrow \sim\alpha$
i. $(\alpha \wedge \beta \Rightarrow \gamma) \Leftrightarrow (\alpha \wedge \sim\gamma \Rightarrow \sim\beta)$
j. $(\alpha \Rightarrow (\beta \Rightarrow (\beta \Rightarrow \alpha)))$
k. $\alpha \Rightarrow (\beta \Leftrightarrow \beta)$
l. $(\alpha \Rightarrow (\beta \Leftrightarrow \gamma)) \Leftrightarrow (\alpha \Rightarrow (\sim\beta \Leftrightarrow \sim\gamma))$
m. $((\alpha \wedge \beta) \wedge (\alpha \Rightarrow \gamma)) \Rightarrow (\gamma \wedge \beta)$
n. $((\alpha \vee \beta) \wedge (\alpha \Rightarrow \gamma)) \Rightarrow (\gamma \vee \beta)$
o. $(\alpha \Rightarrow (\beta \Rightarrow \gamma)) \Rightarrow (\beta \Rightarrow (\alpha \Rightarrow \gamma))$
p. $(\alpha \Rightarrow (\beta \wedge \gamma)) \Leftrightarrow ((\alpha \Rightarrow \beta) \wedge (\alpha \Rightarrow \gamma))$

3. Prove that each of the wffs in Question 2 is a theorem *with* a formal proof.
4. Prove Theorem 1.7.2
5. Prove that if $\vdash \alpha_1 \Rightarrow \omega, \vdash \alpha_2 \Rightarrow \omega, \ldots, \vdash \alpha_k \Rightarrow \omega$ and $\vdash \sim (\alpha_1 \vee \alpha_2 \vee \cdots \vee \alpha_k) \Rightarrow \omega$, then $\vdash \omega$.
6. Show that each of the following conclusions (C) follows from the given premise..

a. P1: $(p \wedge q) \vee (r \wedge s)$
P2: $p \Rightarrow \sim p$

C: r

b. P1: $(p \vee q) \Rightarrow (r \wedge s)$
P2: $(s \vee t) \Rightarrow u$

C: $p \Rightarrow u$

c. P1: $p \Rightarrow (q \wedge r)$
P2: $(q \vee s) \Rightarrow t$
P3: $p \vee s$

C: t

d. P1: $r \Rightarrow s$
P2: $p \Rightarrow (q \wedge r)$
P3: $\sim(q \wedge s)$

C: $\sim p$

e. P1: $p \Rightarrow (q \vee r)$
P2: $q \Rightarrow s$
P3: $r \Rightarrow \sim p$

C: $p \Rightarrow s$

f. P1: $p \vee (q \Rightarrow r)$
P2: $q \vee r$
P3: $r \Rightarrow p$

C: p

g. P1: $p \wedge q \Rightarrow r$
P2: $p \vee q \Rightarrow u$
P3: $r \Rightarrow \sim u$

C: $p \Rightarrow \sim q$

h. P1: $s \wedge t \Rightarrow \sim p \wedge q$
P2: $(u \wedge \sim v) \vee p$
P3: $u \Rightarrow v \wedge \sim q$

C: $t \Rightarrow \sim s$

i. P1: $q \Rightarrow \sim p$
P2: $\sim q \wedge \sim s \Rightarrow \sim r$
P3: $\sim r \wedge \sim u \Rightarrow \sim t$
P4: $s \Rightarrow q$

C: $\sim t \vee (p \Rightarrow u)$

j. P1: $\sim q \vee p$
P2: $\sim r \vee (s \wedge \sim p)$
P3: $\sim t \vee r \vee q$

C: $\sim s \Rightarrow (\sim t \vee p)$

k. P1: $(r \vee s) \Rightarrow (p \vee q)$
P2: $(t \Rightarrow u) \Rightarrow r$
P3: $t \Rightarrow (u \wedge s)$

C: $\sim p \Rightarrow q$

l. P1: $p \Rightarrow \sim q$
P2: $t \Rightarrow p \vee r$
P3: $(\sim t \wedge \sim u) \Rightarrow \sim s$
P4: $r \Rightarrow p$

C: $\sim q \vee (s \Rightarrow u)$

m. P1: $p \vee q$
P2: $p \Rightarrow r$
P3: $q \Rightarrow s$

C: $r \vee s$

n. P1: $(p \Rightarrow q) \wedge (r \Rightarrow s)$
P2: $(q \Rightarrow t) \wedge (s \Rightarrow u)$
P3: $\sim (t \wedge u)$
P4: $p \Rightarrow r$

C: $\sim p$

7. Verify that each of the following arguments is valid with a formal proof.

a. If Joe is a mathematician, then he is logical.
If Joe is romantic, then he likes soap opera.
If Joe is logical, then he does not like soap opera.
Hence, if Joe is a mathematician, then he is not romantic.

b. If Frank does not live in Italy, then he does not speak Italian.
Frank does not own a motocycle.
If Frank lives in Italy, then he owns a scooter.
Either Frank speaks Italian or he owns a motocycle.
Hence, Frank owns a scooter.

c. The contract is satisfied if and only if the building is completed by November 30.
The electrical subcontractor completes his work by November 10 if and only if the building is completed by November 30.
The bank loses money if and only if the contract is not satisfied.
Hence, the electrical subcontractor completes his work by November 10 if and only if the bank does not lose money.

d. If the casino is shut down or a gambling tax is imposed, then tourism will decline and the city will suffer. If tourism declines, then the city will be a safer place to live. The city is not a safer place to live. Therefore, the casino is not closed.

e. If Mario leads his team to win the cup, he will be arrogant. If he is arrogant, then he is not a good team leader. If he fails in leading his team to win the cup, he will lose his leadership. He is not a good team leader if he loses his leadership. If he is not a good team leader, then he should retire. Therefore, Mario should retire.

f. Frank is on his way to the Wonder Land. He loses his way and reaches a junction of three streets: Gold Street, Park Street and High Street. There is guardian on each street. Frank asks each of them for direction.
The Gold Street guardian answers: "The Gold Street does not lead you to the Wonder Land."
The Park Street guardian answers: "The Park Street does not lead you to the Wonder Land."
The High Street guardian answers: "The Park Street leads you to the Wonder Land."

Frank knows the following facts:

"Only one of the three streets leads to the Wonder Land, the other two does not."

"Exactly one of the three answers is true."

Use propositional logic to prove that the Gold Street leads to the Wonder Land, and the Park Street and High Street do not.

g. Andrew, Betty and Carlo come to the party if and only if David does not come.
If neither Andrew nor Betty come, then David comes only if Carlo comes.
Betty does not come to the party.
Hence, Andrew or Carlo comes to the party.